

What you could have in 2047 is 4D communication technology. It promises to make life more comfortable and exciting. Imagine having 4D maps to help autonomous (driverless) vehicles navigating roads or flight paths, ensuring zero collision and the fastest commute. Or your own robotic exoskeleton—a whole body suit a la Iron Man—which responds to your thoughts and uses the power of machines to augment physical capabilities. Shaky hand, did you say? Your exoskeleton will hold your glass for you so that you do not spill anything. And that's only a simple task, it can follow much more complex commands. A quantum leap in computing, communications and extraordinary human creativity have led to a convergence of technologies that is making possible what was once considered impossible.

Welcome to the Brave All-New World. Prime Minister Narendra Modi, in his Independence Day address this year, called the next 24 years *Amrit Kaal* (auspicious period) and urged citizens to collectively work towards transforming India into a developed nation by 2047. However, the definition of a developed India@100 is vastly different from the conventional yardstick currently employed to categorise a developed country. Today, it not only means a country that has sustainable economic growth with a robust and efficient transport and communications infrastructure. It necessarily entails a well-rounded human development index, reflecting a uniformly high standard of living with a healthy and well-educated citizenry. We must also aim for high per capita incomes averaging $20,000 (nearly Rs 16.7 lakh) annually or more. India currently has a per capita income of $2,400 (Rs 2 lakh), which, according to the World Bank standard, puts us in the class of lower-middle-income economies, where the per capita income ranges from $1,136 to $4,465 (approximately Rs 94,400-Rs 3.7 lakh). We are, therefore, in the same boat as Bangladesh, Sri Lanka and Pakistan. The per capita income in China, whose GDP at $18 trillion is nearly six times India's $3.4 trillion and whose population has now become second to ours, is $12,800 (approx. Rs 10.6 lakh). That places it in the upper-middle-income category but not in the league of developed nations.

What works in India's favour is what PM Modi identified as a trinity of Ds—demography, democracy and diversity. From being the fifth-largest economy in the world, he is confident that India will become the third-largest before 2030, going past Japan and Germany. Around this time, China may overtake the US' GDP of an estimated $28 trillion and emerge as the world's largest economy. But as both India and China demonstrate, the sheer size of an economy does not determine its status as a developed country.

So, what will it take for India to become a truly developed country when it completes 100 years of Independence on August 15, 2047? There is no way we can achieve this exalted status with business as usual. Instead, the country will need a revolution in 10 key sectors—Energy, Transport, Communications, Agriculture, Defence, Health, Education, E-governance, E-commerce and Computing. Driven by rapid technological innovations, each of these major fields is undergoing dramatic changes. If India does not keep abreast of, and adapt to, these developments, it will be left far behind, with exploitation and deprivation as its permanent fate.

Let's start with Communications. India can take pride in the fact that its average cost of Rs 14 for 1GB of mobile internet data is among the lowest in the world. Users in the US, in comparison, pay 33 times more and those in the UK 4.5x what we do. Also to its credit, India has moved quickly to be in lockstep with advanced nations in the introduction of 5G telecom networks. However, the real test will come with next generation telephony—6G. It is designed